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Callus Induction and phytochemical screening of *Tinospora cordifolia* (Willd.) Miers ex Hook. F. and Thoms.

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Abstract : It is one of the most versatile rejuvenating shrub also known as 'Giloya' in Indian vernacular having many therapeutic applications. Various parts of the plant are being prescribed in Ayurveda and other systems of medicine as a monoherbal or polyherbal preparation. The pharmaceutical significance of this plant is mainly because of the leaves, barks and roots contain various bioactive compounds such as alkaloids, glycosides, lactones, steroids, polysaccharides and aliphatic compounds having various medicinal importances; a few important applications are viz. immunomodulatory or immunostimulatory, anti-neoplastic, anti-oxidant, gastrointestinal and hepatoprotection, and side effects prevention of the cancer chemotherapy. The present investigations were carried out with a view to standardize a protocol for callus induction and to substantiate medicinal properties through phytochemical screening. The nodal explants showed 100% callusing in MS medium incorporated with NAA (6mg/L). In another combination of BAP (0.5mg/L) with IBA (0.5mg/L) and 2-4-D (0.5 mg/L) callusing was seen on leaf explant (8%), while on nodal segment 100% callusing was observed. Different parts of the plant specially stem and leaves of *T. cordifolia* are very rich in different types of alkaloids, such as saponins, tannins, alkaloids, steroids, terpenoids, glycosides, resins, phlobatannins, resins, flavonoids, cardioglycosides and alkaloides..

Key Words: *Tinospora cordifolia*, callus, immunomodulator, anti-neoplastic, phytochemical

INTRODUCTION

Tinospora cordifolia (Willd.) Miers ex Hook. F. and Thoms. (Family: Menispermaceae) commonly known, as "Amrita" is a large deciduous climbing shrub found throughout tropical Indian subcontinent. The term Amrita signifies its ability to impart youthfulness, vitality and longevity. The pharmaceutical significance of this plant is mainly because of the leaves, barks and roots, which contain various bioactive compounds such as alkaloids, glycosides, lactones, steroids, polysaccharides and aliphatic compounds having various medicinal importance viz. immunomodulatory or immunostimulatory, anti-tumor, cognition, anti-inflammatory, anti-neoplastic, anti-hyperglycemia, anti-hyperlipidemia, anti-oxidant, anti-tuberculosis, gastrointestinal and hepatoprotection, anti-osteoporotic, anti-angiogenic, anti-malarial, anti-allergic and side effects prevention of the cancer chemotherapy. The bitter priniciples present in the leaves, stems, roots and barks viz. tinosporine, tinosporide, tinosporaside, cordifolide, cordifol, berberine, cordifolioside A, B. C, amritosides A, B, C, and columbin which act as therapeutic agents and play vital role in many therapeutic applications.[1] Tissue culture technique holds much potential as they can be employed as an effective tool for enhancement of bioactive compounds through biotransformation. Therefore, there is an urgent need to establish an effective and efficient protocol for culture of plant parts utilized in traditional system of medicine against different types of ailments. With this purpose attempts have been made to establish with this perpose attempts have been made to undertake phytochemical screening and callus induction

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MATERIALS AND METHODS

The plant material of the experimental plant, *T. cordifolia* was obtained from the Medicinal Garden of the Department for phytochemical screening and callus induction.

PHYTOCHEMICAL SCREENING

The plant samples (leaves and stems) were air-dried and ground into uniform powder.The methanolic extract of each sample was prepared by soaking 10 gm of dried powdered samples in 100 ml of methanol. The extracts were filtered after continuous soaking and shaking for 72 hours using Whatman filter paper No 1. The chemical tests adopting standard procedures were carried out either on the methanolic extract or extract powder so as to identify the secondary metabolites.as constituents as described by Sofowara (1993), Trease and Evans (1989) and Harborne (1973).

Test for tannins: About 0.5 gm of the dried powdered samples was boiled in 20 ml of water in a test tube and then filtered. A few drops of 0.1% ferric chloride was added and observed for browrish green or a blue-black colouration.

Test for phlobatannins: Deposition of a red precipitate when a methanolic extract of each plant sample was boiled with 1% aqueous hydrochloric acid was taken as evidence for the presence of phlobatinins.

Test for saponin: About 2 gm of the powdered sample was boiled in 20 ml of distilled water in a water bath and filtered. 10ml of the filtrate was mixed with 5 ml of distilled water and shaken vigorously for a stable persistent froth. The frothing was mixed with 3 drops of olive oil and shaken vigorously, then observed for the formation of emulsion.

Test for flavonoids: Three methods were used to determine the presence of flavonoids in the plant sample. 5 ml of dilute ammonia solution was added to a portion of the aqueous filtrate of each plant extract followed by addition of concentrated H_2SO_4. A yellow colouration observed in each extract indicated the presence of flavonoids. The yellow colouration disappeared on standing. A few drops of 1% aluminium solution were added to a portion of each filtrate. A yellow colouration was observed indicating the presence of flavonoids. A portion of the powdered plant sample was in each case heated with 10 ml of ethyl acetate over a steam bath for 3 min. The mixture was filtered and 4 ml of the filtrate was shaken with 1 ml of dilute ammonia solution. A yellow colouration was observed indicating a positive test for flavonoids.

Test for steriods: Two ml of acetic anhydride was added to 0.5 gm ethanolic extract of each sample with 2 ml H_2S0_4. The colour changed from violet to blue or green in some samples indicating the presence of steroids.

Test for terpenoids (Salkowski test): Five ml of each extract was mixed in 2 ml of chloroform, and concentrated H_2S0_4 (3 ml) was carefully added to form a layer. A reddish brown colouration of the inter face was formed to show positive results for the presence of terpenoids.

Test for cardiac glycosides (Keller-Killani test): Five ml of each extracts was treated with 2 ml of glacial acetic acid containing one drop of ferric chloride solution. This was underlayed with 1 ml of concentrated sulphuric acid. A brown ring of the interface indicates a deoxysugar characteristic of cardenolides. A violet ring may appear below the brown ring, while in the acetic acid layer, a greenish ring may form just gradually throughout thin layer.

CALLUS INDUCTION

Six-month-old twines of healthy plants of *T. cordifolia* were collected from the departmental garden of this institute. Explants were prepared from the nodal portion of stems, which were thoroughly washed with tap water to remove soil and other superficial contamination, and then allowed to stay for 3–4 h in double distilled water to facilitate the leaching out of phenolics and a characteristic gummy substance of polysaccharide from the explants.

The explants were sterilized with 70% ethanol (5 minutes) followed by 0.1% $HgCl_2$ (3 minutes) treatment. The explants were washed with double distilled water 5 to 6 times. The sterile explants of *Tinospora cordifolia* were used for inoculation in Murashige and Skoogs (MS) medium with single or different combination of Indole butyric acid (IBA), 2, 4-diclorophenoxy acetic acid (2-4-D), Benzyl amino purine (BAP) and Naphthelene-1- acetic acid (NAA). The inoculated culture tubes were maintained in culture room under controlled conditions: temperature 25 ± 3°C; light period - 16 hours light and 8 hours dark; light intensity - 1500- 3000 lux and relative humidity 60%.

RESULTS

Phytochemical screening showed that The leaf and stem of *Tinospora cordifolia* contain saponins, tannins, alkaloids, steroids, terpenoids, glycosides, resins, phlobatannins, resins, flavonoids, cadioglycosides and alkaloides. Results show presence or absence of secondary metabolites. Dark colour or precipitation is indication of high concentrations of the compounds in question (Table 3).

In order to induce callusing, leaf and nodal segments of *Tinospora cordifolia* were incubated in MS medium incorporated either with auxins (2,4-D; NAA) and cytokinin (BAP) alone or their combinations. The concentrations of different auxins and BAP were 1,2,,3,4,5,6,7 mg/L Callus induction was seen in MS supplemented with different concentrations of NAA (2,3,6 mg/L). The highest frequency of callusing (100%) was seen at concentration of 6 mg/L NAA, whereas 90% , 75%, 60% and 8% callusing were seen at 5, 4, 3, 2mg/L NAA, respectively. However, no callusing was found at 1mg/L and 7 mg/L of NAA. Interestingly, rooting was also seen together with callus at 6mg/L of NAA. Other hormones, 2,4-D and BAP when incorporated in MS medium at different concentrations ,as mentioned above, no callusing was seen (Table 1; Fig 1A). The callus was profuse and creamy and sticky in nature.

MS medium in combination of BAP (1mg/L) and NAA (2.5mg/L) showed no callusing on leaf, while nodal segment showed 60% callusing (Table 2). In another combination of BAP (0.5mg/L) with IBA (0.5mg/L) and 2-4-D (0.5 mg/L) callusing was seen on leaf explant (8%), while on nodal segment 100% callusing was observed (Figs.1 B, C). In third combination of hormones BAP (1.0 mg/L) with IBA (1.0 mg/L) and 2-4-D (1.0 mg/L leaf explant showed no callusing, while nodal segment gave 70% callusing (Table 2; Figs. 1 D,E). The callus formation was observed after 25 and 30 days of inoculation, showing compact and sticky nature. After 40 days callus started browning and gradually dies.

DISCUSSION

Callus induction and regeneration are the basic protocol for any advance studies, like qualitative and quantitative improvement of the plant species. Micropropagation is another area of plant tissue culture for multiplication/ biomass enhancement of the plant species which are endangered or for scaling up the biomass for large quantity extraction of bioactive compounds. In the present investigation callus was obtained from leaf and stem.

Different parts of the plant specially stem and leaves of *T. cordifolia* are very rich in different types of alkaloids, such as saponins, tannins, alkaloids, steroids, terpenoids, glycosides, resins, phlobatannins, resins, flavonoids, cadioglycosides and alkaloides. Presences of these compounds suggest that the plant is a potential multipurpose medicinal plant. Presence of different types of bioactive compound in the same plant have also been reported by different worker.[2,3,4,5,6,7]

All plants produce chemical compounds as part of their normal metabolic activities. However, some of the bioactive compounds are of medicinal properties.[8,9] The bioactive compounds of *T. cordifolia* has been reported by numbers of scientists[10,11,12] and also medicinal properties have been confirmed.[13,14,15,16]

CONCLUSION

Tinospora cordifolia is in Ayurvedic medicine and also in other traditional system of medicines. It is highly rich in secondary metabolites of therapeutic potentials. Moreover, its wide application is proved on the basis of its bioactive molecules and pharmacological evaluation by various modern tests. The plant species is good candidate for the elicitation/ bioconversion of active compounds as shown by preliminary studies of callus induction and phytochemical screening from stem and leaf explants.

Table1: Effect of different concentrations of auxins on nodal explant of *Tinospora cordifolia* in MS medium

S. No.	Hormone	Concentration (mg/L)													
		1.0		2.0		3.0		4.0		5.0		6.0		7.0	
		Degree of Callus	% callus	Degree of callus	% callus	Degree of callus	% callus	Degree of callus	% callus	Degree of callus	% callus	Degree of callus	% callus	Degree of callus	% callus
1.	2,4-D	_	_	_	_	_	_	_	_	_	_	_	_	_	_
2.	NAA	_	_	+	08	++	60	++	75	++	90	++ R	100	_	_
3.	BAP	_	_	_	_	_	_	_	_	_	_	_	_	_	_

(-): Nil; (+): Good; (++): Very good; R: Root

Table 2: Effect of combination of auxins and cytokinins on leaf and nodal explants of *Tinospora cordifolia* in MS medium

Explant	BAP(1mg/L)+ NAA(2.5mg/L)		BAP(0.5mg/L)+ IBA(0.5mg/L) + 2,4-D (0.5mg/L)		BAP (1.0mg/L) + IBA(1.0mg/L) + 2,4-D(1.0 mg/L)	
	Degree of callusing	% callus	Degree of callusing	% callus	Degree of callusing	% callus
Leaf	_	_	++	8%	_	_
Nodal segment	+	60	+++	100%	+++	70%

(-): Nil; (+): Good; (++): Very good; (+++): Excellent

Table 3: Phytochemical screening of secondary metabolites in *Tinospora cordifolia*

S. No	Plant part	Saponin	Tannin	Steroids	Terpenoids	Glycosides	Resins	Phlobatannin	Flavanoidds	Cardiac Glycosides	Alkaloids
1	Stem	+	++	++	++	+	++	++	+	+	+++
2	Leaf	+	+	+	+	+	+	--	++	+	+

(+++): appreciable amount; (++): moderate amount; (+): trace; (-): absent

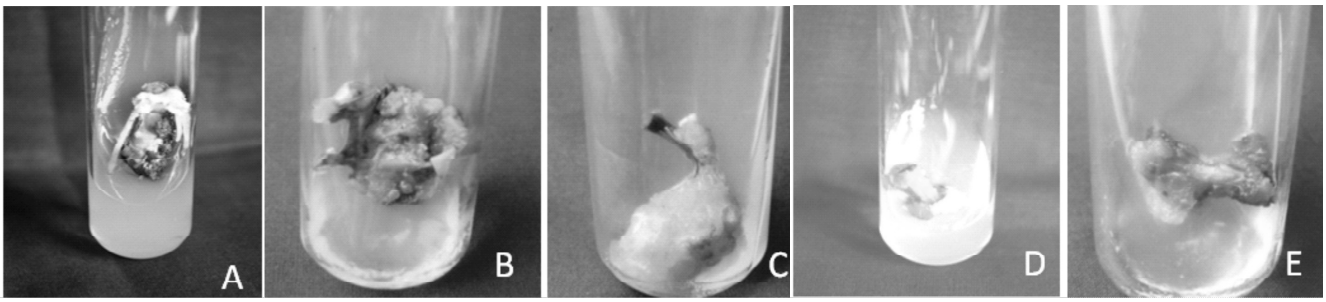


Fig. Callus induction in explantsat different concentration of growth homone in MS medium.
A: Stem MS+NAA (6mg/L); B: Leaf IBA+BAP+2,4-D (0.5mg/L each); C: Stem IBA+BAP+2,4-D (0.5mg/L each); D: Stem IBA+BAP+2,4D (1mg/L each); E: Stem IBA+BAP+2,4-D (1mg/L each)

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